

Practical-11

AIM: Develop an Android application that uses Services to perform background tasks.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical11_layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical11">
<Button
    android:id="@+id/Practical11_start_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start the service"

app:layout_constraintBottom_toTopOf="@id/Practical11_stop_btn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
<Button
    android:id="@+id/Practical11_stop_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Stop the service"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Practical11_start_btn" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Service.java

```
package com.mk.madpractical;

import android.app.Service;
import android.content.Intent;
import android.os.Binder;
import android.os.Handler;
import android.os.Looper;
```

```
import android.os.IBinder;
import android.widget.Toast;

public class Practical11Service extends Service {

    private Thread thread;
    private boolean isServiceRunning;

    public Practical11Service() {
    }

    @Override
    public IBinder onBind(Intent intent) {
        return new Binder() {
            Practical11Service get_service() {
                return Practical11Service.this;
            }
        };
    }

    @Override
    public void onCreate() {
        super.onCreate();
        isServiceRunning = true;

        // Create and start a new thread
        thread = new Thread(new Runnable() {
            @Override
            public void run() {
                while (isServiceRunning) {
                    new
                    Handler(Looper.getMainLooper()).post(new Runnable() {
                        @Override
                        public void run() {
                            Toast.makeText(Practical11Service.this,
                                "Service is running", Toast.LENGTH_SHORT).show();
                        }
                    });
                }
            }
        });

        @Override
        public int onStartCommand(Intent intent, int flags, int
```

```

startId) {
    if (!thread.isAlive()) {
        thread.start(); // Start the thread
    }
    return START_STICKY; // Keep the service running
}

@Override
public void onDestroy() {
    super.onDestroy();
    isServiceRunning = false; // Stop the loop
    if (thread != null && thread.isAlive()) {
        thread.interrupt(); // Safely stop the thread
    }
}
}

```

Practical11.java

```

package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical11 extends AppCompatActivity {
    Button start,stop;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

```

```

        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical11);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical11_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        start = findViewById(R.id.Practical11_start_btn);
        stop = findViewById(R.id.Practical11_stop_btn);
        Intent service = new
        Intent(this,Practical11Service.class);
        View.OnClickListener listener = new
        View.OnClickListener() {
            @Override
            public void onClick(View v) {

                try {
                    if (v.equals(start)){
                        startService(service);
                    }else {
                        stopService(service);
                    }
                } catch (Exception e) {
                    throw new RuntimeException(e);
                }
            }
        };
        start.setOnClickListener(listener);
        stop.setOnClickListener(listener);
    }
}

```

